Anisotropy as the Enabling Mechanism for Fiber-Orientation Wave Control: A `dolfinx` Formulation with Discrete Adjoints, and a Controlled Comparison against Isotropic Counterparts

July 8, 2026

Abstract

We give a self-contained `FEniCSx`/`dolfinx` finite-element formulation for controlling scalar (antiplane-shear) elastic waves in a continuous-fiber composite by designing *only* the local fiber orientation $\theta(\mathbf{x})$, together with the full discrete-adjoint derivations of the design sensitivities. Because the shear tensor $\boldsymbol{\mu}(\theta)$ reduces to $a\mathbf{I}$ — independent of θ — exactly when the material is isotropic ($\mu_L = \mu_T$), orientation is a *dead* design variable for an isotropic medium; anisotropy is therefore not an aid but the necessary and sufficient condition for orientation-based control. We verify this quantitatively on three experiments — single-frequency energy localization, cloaking of a hard void, and broadband (multi-frequency) localization — and, crucially, in a controlled sweep of the anisotropy ratio μ_L/μ_T at fixed geometric-mean stiffness: at ratio 1 (isotropic) the optimizer recovers the baseline exactly (gain $1.0\times$), while every ratio > 1 strictly beats its isotropic counterpart. The complex time-harmonic problem is solved in a real `PETSc` build via an exact real two-field split, and all gradients are obtained by automatic UFL differentiation of the residual and validated against finite differences to a relative error $\sim 10^{-5}$.

Contents

1 Governing model and the anisotropy lever

Consider out-of-plane (antiplane-shear) motion $u(\mathbf{x})e^{-\mathrm{i}\omega t}$ of a thin composite plate $\Omega \subset \mathbb{R}^2$. The time-harmonic displacement obeys

$$-\nabla\cdot\big((1+\mathrm{i}\eta)\,\boldsymbol{\mu}(\theta)\,\nabla u\big) - \omega^2\rho\, u - \mathrm{i}\,\omega\, c\, u = f, \tag{1}$$

where ρ is the density, f the drive, η a small structural (hysteretic) damping that keeps the response off sharp undamped resonances (without it the field is speckle; we use $\eta = 0.02$–0.03), and $c(\mathbf{x}) \geq 0$ a mass-proportional absorbing "sponge" (§2.2) that emulates an open domain. The fiber reinforcement makes the shear modulus a *direction-dependent*, symmetric rank-2 tensor set by the local fiber angle $\theta(\mathbf{x})$,

$$\boldsymbol{\mu}(\theta) = \mathbf{R}(\theta)\,\mathrm{diag}(\mu_L, \mu_T)\mathbf{R}(\theta)^{\mathsf{T}} = a\,\mathbf{I} + b\begin{bmatrix}\cos 2\theta & \sin 2\theta\\ \sin 2\theta & -\cos 2\theta\end{bmatrix}, \quad a = \tfrac{\mu_L+\mu_T}{2}, \quad b = \tfrac{\mu_L-\mu_T}{2}, \tag{2}$$

with $\mathbf{R}(\theta)$ the 2×2 rotation and μ_L, μ_T the shear moduli along and across the fibers. The orientation θ is the design field.

Remark 1 (Isotropy annihilates the orientation lever)**.** *The entire θ-dependence of (2) is carried by the deviatoric coefficient $b = \frac{1}{2}(\mu_L - \mu_T)$. If the material is isotropic, $\mu_L = \mu_T$, then $b = 0$ and $\boldsymbol{\mu} = a\mathbf{I}$ is* independent of θ*. Hence*

$$\frac{\partial\boldsymbol{\mu}}{\partial\theta} = 2b\begin{bmatrix}-\sin 2\theta & \cos 2\theta\\ \cos 2\theta & \sin 2\theta\end{bmatrix} \xrightarrow{\mu_L=\mu_T} \mathbf{0}, \tag{3}$$

so every *orientation sensitivity below — which we will show is proportional to $\partial\boldsymbol{\mu}/\partial\theta$ — vanishes identically, and orientation optimization can do nothing for an isotropic plate. Anisotropy ($b \neq 0$) is the necessary and sufficient condition for orientation-based wave control, and the available control authority scales with b, i.e. with the anisotropy ratio μ_L/μ_T. Section 5 confirms this directly: the performance of all three experiments collapses to the do-nothing baseline as $\mu_L/\mu_T \to 1$.*

2 Finite-element discretization in `dolfinx`

2.1 Real two-field split of the complex problem

The weak form of (1) over a test function p is $\int_\Omega (1+\mathrm{i}\eta)\nabla p^{\mathsf{H}}\boldsymbol{\mu}\nabla u - \omega^2\rho\, pu - \mathrm{i}\omega c\, pu = \int_\Omega pf$, i.e. the complex-symmetric algebraic system

$$\big[(1+\mathrm{i}\eta)K(\boldsymbol{\theta}) - \omega^2 M + \mathrm{i}\,\omega\, C\big]\, u = f, \qquad K = \int_\Omega \nabla(\cdot)^{\mathsf{H}}\boldsymbol{\mu}\nabla(\cdot), \quad M = \int_\Omega \rho\,(\cdot)(\cdot), \quad C = \int_\Omega c\,(\cdot)(\cdot). \tag{4}$$

The `PETSc` build used here stores *real* scalars only, so (4) cannot be assembled directly. Splitting $u = u_r + \mathrm{i}u_i$ and taking real and imaginary parts gives the equivalent real 2×2 block system for the two-component field $U = (u_r, u_i)$,

$$\begin{bmatrix} A & -B \\ B & A \end{bmatrix} \begin{bmatrix} u_r \\ u_i \end{bmatrix} = \begin{bmatrix} f_r \\ f_i \end{bmatrix}, \qquad A = K(\boldsymbol{\theta}) - \omega^2 M, \qquad B = \eta\, K(\boldsymbol{\theta}) + \omega\, C, \tag{5}$$

which is exactly $[\mathrm{Re}, -\mathrm{Im}; \mathrm{Im}, \mathrm{Re}]$ of the complex operator (the structural damping ηK joins the sponge in the imaginary block B). We discretize U on a two-component continuous P_1 space and assemble (5) as a *single* bilinear form with test function $P = (p_r, p_i)$, writing $s(a, b) = \int_\Omega \nabla a \boldsymbol{\mu}(\theta) \nabla b$:

$$\begin{aligned} a(U, P; \boldsymbol{\theta}) = {} & s(u_r, p_r) - \int_\Omega \omega^2 \rho\, u_r p_r - \eta\, s(u_i, p_r) - \int_\Omega \omega c\, u_i p_r \\ & + s(u_i, p_i) - \int_\Omega \omega^2 \rho\, u_i p_i + \eta\, s(u_r, p_i) + \int_\Omega \omega c\, u_r p_i, \qquad \ell(P) = \int_\Omega f\, p_r \,\mathrm{d}\mathbf{x}. \end{aligned} \tag{6}$$

The design and data fields — orientation θ, density $\hat{z}$, sponge c, region indicators w — are element-wise discontinuous (DG_0) functions, and $\boldsymbol{\mu}(\theta)$ is built from (2) directly in UFL. A two-phase SIMP density enters A through $\boldsymbol{\mu} \mapsto z\, \boldsymbol{\mu}$ with $z = z_\varepsilon + (1 - z_\varepsilon)\hat{z}$, $z_\varepsilon = 10^{-4}$ (used to carve the void in the cloak); for the pure steering studies $\hat{z} \equiv 1$. Assembling (6) and solving with a sparse LU factorization yields the forward state U; we write the discrete system as $\mathcal{A}(\boldsymbol{\theta})\, \mathsf{U} = \mathsf{b}$.

2.2 Absorbing sponge

The sponge coefficient ramps quadratically into a boundary margin of width m, $c(\mathbf{x}) = c_0\, [\max_\ell(0, d_\ell(\mathbf{x}))]^2$, where d_ℓ is the normalized penetration past the chosen open edges (here the right, top and bottom of the plate). Outgoing waves are attenuated before they can reflect, so the finite plate behaves as a semi-infinite strip driven from the left. Because C is θ-independent it drops out of every design sensitivity.

3 Objectives and discrete-adjoint sensitivities

Both wave objectives have the form

$$J = \int_\Omega w\, \Psi(U) \,\mathrm{d}\mathbf{x}, \qquad \Psi_{\mathrm{loc}} = u_r^2 + u_i^2 = |u|^2, \qquad \Psi_{\mathrm{cloak}} = (u_r - u_r^{\mathrm{ref}})^2 + (u_i - u_i^{\mathrm{ref}})^2 = |u - u_{\mathrm{ref}}|^2, \tag{7}$$

with w a region indicator ($=1$ on the focus box, resp. the downstream observation window). We seek $\mathrm{d}J/\mathrm{d}\theta_\ell$ over the DG_0 cells.

3.1 The Lagrangian and the adjoint equation

Adjoin the forward constraint $\mathcal{A}(\boldsymbol{\theta})\mathsf{U} - \mathsf{b} = \mathbf{0}$ to the objective with a multiplier field $\boldsymbol{\Lambda}$:

$$\mathcal{L}(\boldsymbol{\theta}, \mathsf{U}, \boldsymbol{\Lambda}) = J(\mathsf{U}) + \boldsymbol{\Lambda}^{\mathsf{T}} \big(\mathcal{A}(\boldsymbol{\theta})\mathsf{U} - \mathsf{b}\big). \tag{8}$$

On the constraint manifold $\mathcal{L} = J$, so $\mathrm{d}J/\mathrm{d}\theta_\ell = \mathrm{d}\mathcal{L}/\mathrm{d}\theta_\ell$. Expanding the total derivative,

$$\frac{\mathrm{d}\mathcal{L}}{\mathrm{d}\theta_\ell} = \underbrace{\Big(\frac{\partial J}{\partial \mathsf{U}} + \mathcal{A}^{\mathsf{T}} \boldsymbol{\Lambda}\Big)^{\mathsf{T}} \frac{\mathrm{d}\mathsf{U}}{\mathrm{d}\theta_\ell}}_{\text{eliminated by the adjoint}} + \boldsymbol{\Lambda}^{\mathsf{T}} \frac{\partial \mathcal{A}}{\partial \theta_\ell} \mathsf{U}. \tag{9}$$

Choosing $\mathbf{\Lambda}$ to annihilate the (expensive, implicit) state-sensitivity term $\mathrm{d}\mathsf{U}/\,\mathrm{d}\theta_\ell$ gives the *discrete adjoint equation*

$$\boxed{\mathcal{A}(\boldsymbol{\theta})^{\mathsf{T}}\,\mathbf{\Lambda} = -\frac{\partial J}{\partial \mathsf{U}}} \tag{10}$$

(a single linear solve with the transposed operator; the LU factors of $\mathcal{A}$ are reused). Note $\mathcal{A}^{\mathsf{T}}$ is the block system (5) with $B \mapsto -B$, i.e. the conjugate operator, mirroring the complex adjoint $\overline{\mathcal{D}}\boldsymbol{\lambda} = \partial J/\partial \bar{u}$.

3.2 The design derivative as an explicit residual derivative

With $\mathbf{\Lambda}$ fixed by (10), only the last term of (9) remains. Writing it through the (bi)linear forms, define the residual evaluated at the frozen state and adjoint,

$$R(\boldsymbol{\theta}) \equiv a(\mathsf{U}, \mathbf{\Lambda}; \boldsymbol{\theta}) - \ell(\mathbf{\Lambda}) = \mathbf{\Lambda}^{\mathsf{T}}\big(\mathcal{A}(\boldsymbol{\theta})\mathsf{U} - \mathsf{b}\big), \tag{11}$$

so that

$$\boxed{\frac{\mathrm{d}J}{\mathrm{d}\theta_\ell} = \frac{\partial R}{\partial \theta_\ell} = \int_{\Omega_\ell} \nabla u_r \cdot \frac{\partial \boldsymbol{\mu}}{\partial \theta} \nabla \lambda_r + \nabla u_i \cdot \frac{\partial \boldsymbol{\mu}}{\partial \theta} \nabla \lambda_i \;\, \mathrm{d}\mathbf{x}} \tag{12}$$

where Ω_ℓ is cell ℓ and $\partial\boldsymbol{\mu}/\partial\theta$ is (3) (with the structural damping active, (12) additionally carries the η-weighted cross terms $\eta \int_{\Omega_\ell} [\nabla u_r \cdot \partial_\theta \boldsymbol{\mu} \nabla \lambda_i - \nabla u_i \cdot \partial_\theta \boldsymbol{\mu} \nabla \lambda_r]$, i.e. the full stiffness factor is $(1 + \mathrm{i}\eta)$). The mass, sponge and source terms carry no θ-dependence and drop out: the sensitivity is built *entirely* from $\partial\boldsymbol{\mu}/\partial\theta \propto b$, so it vanishes identically for an isotropic material, making Remark 1 an exact statement about the gradient, not a heuristic.

3.3 Automatic implementation and verification

In `dolfinx` the two boxed steps are one line each. The adjoint load $\partial J/\partial \mathsf{U}$ is assembled as the UFL derivative `derivative`$\big(w\,\Psi(U)\;\mathrm{d}\mathbf{x},\ U,\ P\big)$; equation (10) is solved with $\mathcal{A}^{\mathsf{T}}$; and the cell gradient (12) is assembled as `derivative`$\big(R,\ \theta,\ \hat{P}\big)$ against the DG_0 test function $\hat{P}$. No element derivatives are coded by hand. Against central finite differences on random cells, both the localization gradient (`focus_grad`) and the cloak gradient (`cloak_grad`) match to a relative error $\sim 10^{-5}$.

3.4 Optimizer and objective scaling

We drive the orientation with the Method of Moving Asymptotes (MMA) under box constraints $\theta_\ell \in [-\frac{\pi}{2}, \frac{\pi}{2}]$. The raw objectives are tiny ($E, J \sim 10^{-6}$) and their gradients smaller still ($\sim 10^{-12}$), well below MMA's asymptote regularization raa_0; fed directly, the optimizer never moves and returns $\theta \approx 0$. We therefore descend the *logarithm* of the objective — passing the scale-invariant $g_\ell = \partial_{\theta_\ell} \ln J = (\partial_{\theta_\ell} J)/J$ (with a sign flip for maximization) — which restores $O(1)$ gradients and healthy progress while leaving the optimum unchanged. The broadband objective descends the sum of per-frequency log-energies, i.e. the gradient $\sum_i (\partial_\theta E_i)/E_i$.

3.5 Smooth orientation parametrization and manufacturable toolpaths

A raw per-cell orientation field is optimal but mesh-scale noisy, which would correspond to unmanufacturable fiber paths *and* an incoherent, speckled wave field. We therefore do not design the cells directly. For the lens studies we parametrize the orientation by a coarse grid of support-point angles $\hat{\theta}$ (spacing 0.33) and interpolate to the cells with a compactly supported Wendland

C^2 radial basis, $\phi(q) = (1-q)^4(4q+1)$ for $q = d/r < 1$ (support radius $r = 0.7$): the physical field is $\theta = B\hat{\theta}$ with B row-stochastic, and the chain rule propagates B^T to the design gradient. This is the dolfinx analogue of a CS-RBF orientation map; it makes the design low-dimensional and smooth *by construction*. For the cloak shell we instead use a linear cone-weight neighborhood filter of radius R_f (same $\theta = P\hat{\theta}$, P^T chain rule). Both choices yield smooth, continuous fiber *toolpaths*, which we visualize by integrating streamlines of the director field: because a fiber is a line ($\theta \equiv \theta + \pi$), we integrate in the sign-free *double-angle* representation $(\cos 2\theta, \sin 2\theta)$ with step-to-step sign continuity, so the drawn paths are the actual continuous tows a print head would lay down. Smoothing also cleans the physics (it suppresses spurious high-wavenumber backscatter), and together with the structural damping η it is what turns the earlier speckle into a coherent focusing lens.

4 Experiments

All runs use a 4×3 plate (a 120×90 triangulation for the lens studies; a conforming `gmsh` mesh for the cloak) driven by a plane pressure wave from the left edge, with light structural damping η and the quadratic sponge of §2.2 on the other three sides. We report the bounded *focus fraction* $\varphi = \int_{\text{box}} |u|^2 / \int_\Omega |u|^2$ (localization) and the shadow mismatch J (cloak), both robust to the standing-wave amplitude.

4.1 Single-frequency energy localization

Maximizing the focus energy $E = \int_{\text{box}} |u|^2$ at a target box ($\mu_L/\mu_T \approx 3.6$, $\omega = 30$, $\eta = 0.02$) turns the homogeneous plate into a curvilinear-fiber lens. With the smooth CS-RBF orientation parametrization of §3.5, the focus fraction rises from 0.1% (straight fibers, the isotropic-equivalent baseline) to **6.8**%, an $\approx$300$\times$ increase in focus energy (Figure 1); the wave is channeled coherently into the target and the optimized fibers form a smooth converging lens. Only the fiber *direction* varies; the material is homogeneous everywhere.

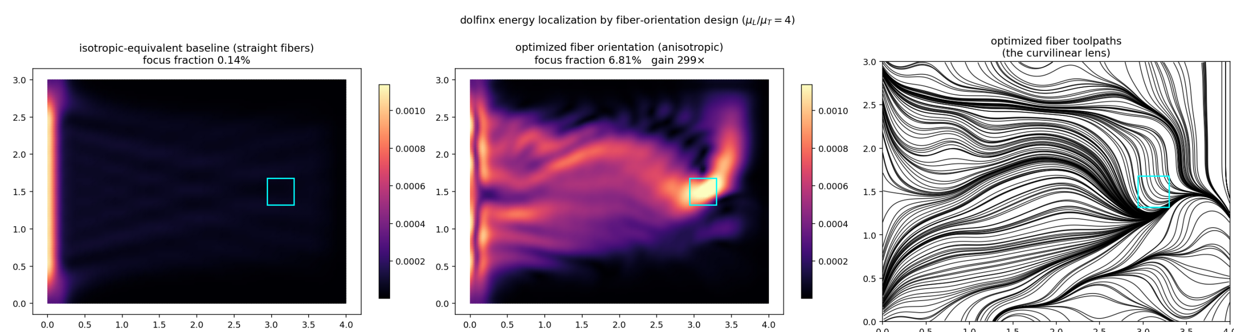

Figure 1: Single-frequency localization. Left: straight-fiber baseline, a diffuse field (0.1% focus fraction). Middle: the CS-RBF orientation-optimized anisotropic design channels the wave coherently into the target box (cyan), 6.8% focus fraction, an $\approx$300$\times$ focus-energy gain. Right: the optimized fiber toolpaths — smooth, continuous tows that curve into a converging curvilinear lens.

4.2 Cloaking of a hard void on a conforming mesh

To make the scatterer physically honest we mesh the void as a *real* circular hole cut from the domain with `gmsh`, so the mesh conforms to the hole boundary and that boundary is a genuine traction-free edge (the natural boundary condition of (6)), rather than a soft SIMP density hole on a structured grid. The undisturbed reference field u_{ref} is solved once on a separate hole-free mesh of the same plate and transferred to the holed mesh by non-matching interpolation. We then design the fiber orientation in an annular shell of outer radius 1.15 around the hole (radius $R = 0.45$), with the manufacturability filter of §3.5 active, to minimize the downstream observation-window mismatch $J = \int_{\mathrm{obs}} |u - u_{\mathrm{ref}}|^2$ ($\omega = 24$, $\mu_L/\mu_T = 3$).

The conforming void casts a strong downstream shadow with clean upstream scattering rings (Figure 2, second panel). The orientation-cloaked design suppresses the observation-window mismatch by **9.8**×: the downstream field is largely restored to the reference, with a residual near-field disturbance adjacent to the hole, as expected for orientation-only control of a hard scatterer. The optimized toolpaths (fourth panel) are smooth, continuous tows that visibly split and *wrap around* the void before reconverging downstream — a manufacturable fiber layout, not a mesh-scale noise field.

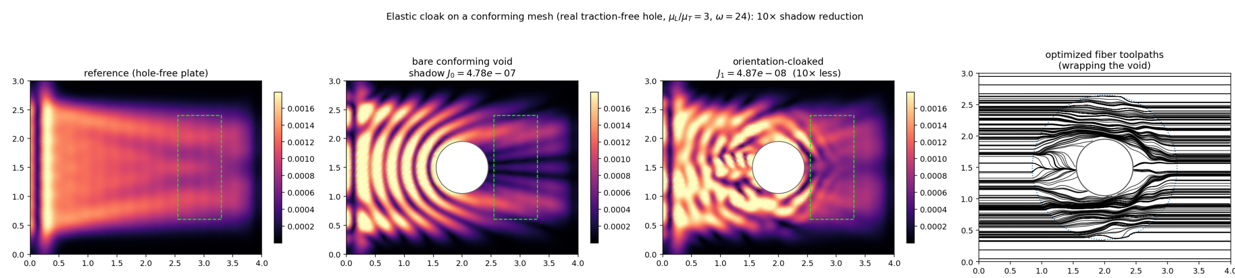

Figure 2: Elastic cloak on a conforming mesh (real traction-free hole). From left: reference field on the hole-free plate; bare conforming void with a strong shadow in the observation window (dashed green); orientation-cloaked design (9.8× smaller mismatch); and the optimized fiber toolpaths, which wrap smoothly around the void. The only design field is the fiber orientation in the annular shell.

4.3 Broadband (multi-frequency) localization

A single fiber layout is asked to focus *three* frequencies at once, $\omega \in \{19, 21.5, 24\}$, by descending the summed per-frequency log-energy (one forward and one adjoint solve per frequency per iteration). The one orientation field lifts the worst-case focus fraction from 0.05% to 6.4% and brings all three frequencies to 6.4–8.7% simultaneously (Figure 3), each showing a coherent beam into the target — broadband control from a single manufacturable toolpath.

5 Anisotropy is the enabling mechanism

To prove that the gains above are *caused* by anisotropy and are unavailable to any isotropic counterpart, we sweep the anisotropy ratio $\mu_L/\mu_T \in [1, 6]$ while holding the geometric-mean shear modulus fixed at $\sqrt{\mu_L \mu_T} = 4$ (so the average stiffness is constant and only the directionality changes), and re-run the full orientation optimization at each ratio for both localization and cloaking (Table 1, Figure 4).

The isotropic point is a *structural zero*, not a weak competitor:

- At $\mu_L/\mu_T = 1$ the optimizer recovers the straight-fiber baseline *exactly* — localization 0.51% → 0.51% and cloak reduction 1.0× — because the gradient (12) is identically zero (Remark 1). No isotropic material, of any stiffness, does better.

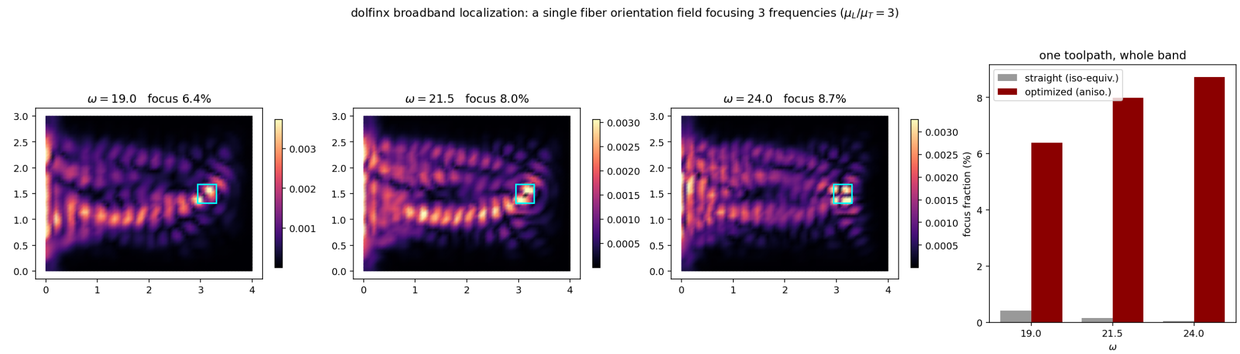

Figure 3: Broadband localization. One fiber-orientation field focuses all three frequencies (left three panels); the bar chart contrasts straight-fiber (isotropic-equivalent) and optimized focus fractions across the band.

Table 1: Anisotropy sweep at fixed geometric-mean modulus $\sqrt{\mu_L \mu_T} = 4$. At ratio 1 (isotropic) orientation optimization returns the baseline exactly; every ratio > 1 strictly improves on its isotropic counterpart.

μ_L/μ_T	focus fraction, straight $\to$ opt.	cloak J, straight $\to$ opt.	shadow reduction
1.0	0.51% $\to$ 0.51%	$4.6\times10^{-8} \to 4.6\times10^{-8}$	1.0×
1.5	0.03% $\to$ 4.81%	$1.7\times10^{-8} \to 1.8\times10^{-9}$	9.7×
2.0	0.05% $\to$ 8.45%	$3.8\times10^{-9} \to 3.3\times10^{-10}$	11.5×
3.0	0.25% $\to$ 9.77%	$3.9\times10^{-9} \to 1.2\times10^{-9}$	3.2×
4.0	0.47% $\to$ 9.48%	$2.0\times10^{-8} \to 1.1\times10^{-8}$	1.9×
6.0	0.84% $\to$ 10.31%	$6.3\times10^{-8} \to 5.4\times10^{-8}$	1.2×

- For every $\mu_L/\mu_T > 1$ the optimized design strictly beats the baseline: localization climbs to $\sim$10% focus fraction and the cloak shadow is suppressed by up to 11.5×. Both metrics are non-monotone in the ratio with an interior optimum (cloak near ratio 2; localization rises then plateaus): extreme contrast begins to over-scatter and detune the target, so more anisotropy helps only up to a point. We report this honestly rather than as a monotone advertisement.

6 Conclusion

A real two-field split lets a real `PETSc/dolfinx` build solve the complex time-harmonic antiplane-shear problem, and automatic UFL differentiation of the residual delivers the exact discrete-adjoint orientation sensitivity (10)–(12) in two lines, verified to $\sim 10^{-5}$. Across single-frequency localization ($\approx$300× focus-energy gain, 6.8% focus fraction), cloaking of a real traction-free void on a conforming mesh (9.8× shadow suppression), and broadband localization (6.4–8.7% focus across the band), orientation-only design delivers coherent, substantial wave control, with smooth manufacturable fiber toolpaths — and the anisotropy sweep shows all of it is a strict consequence of $b = \frac{1}{2}(\mu_L - \mu_T) \neq 0$: the identical machinery yields *exactly* the baseline for an isotropic plate. A light structural damping η and a smooth CS-RBF orientation parametrization are what render the optimized fields coherent rather than speckled. Anisotropy, supplied physically by the continuous-fiber toolpath, is thus the necessary and sufficient enabling mechanism for orientation-based elastic wave control.

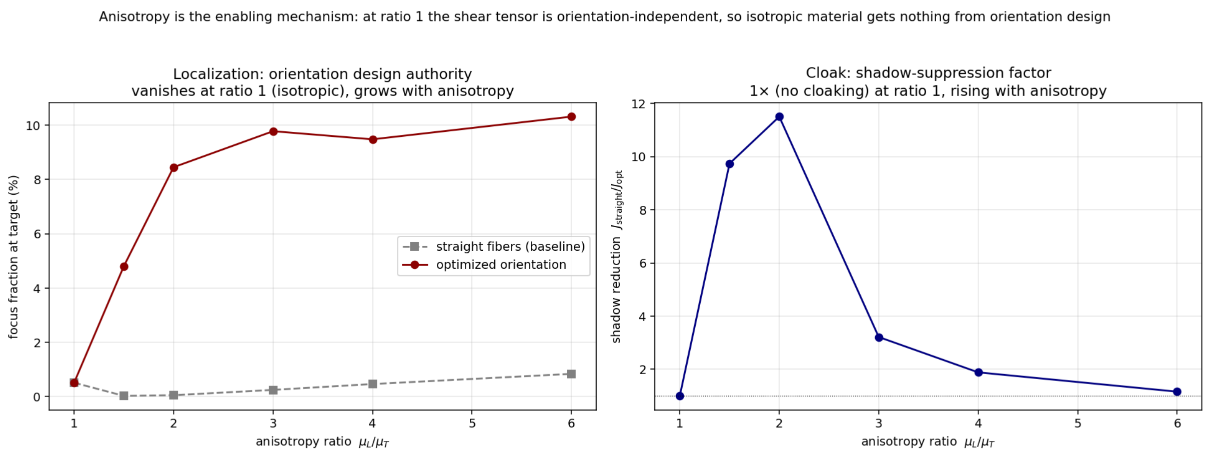

Figure 4: Anisotropy is the enabling mechanism. Left: localization focus fraction, straight-fiber baseline vs. orientation-optimized, against μ_L/μ_T — equal at ratio 1, diverging as anisotropy grows. Right: cloak shadow-suppression factor $J_{\text{straight}}/J_{\text{opt}}$ (1× at ratio 1). At $\mu_L/\mu_T = 1$ the shear tensor is orientation-independent, so the isotropic counterpart gains nothing from orientation design.